

What is an Amplifier in Networking?

An **amplifier** is a device used to increase (boost) the strength of a signal in a network.

 Its main job:

- Take a **weak signal**
- Increase its **power (amplitude)**
- Send it further over the network

⚡ Why do we need an Amplifier?

When data travels through a cable or wireless medium:

- Signal becomes **weak due to distance**
- Noise/interference affects it

👉 So amplifier helps to:

- Extend **transmission distance**
- Maintain **signal strength**

- Electromagnetic interference (EMI)
- Radio frequency interference (RFI)

How Amplifier Works

- Input: Weak signal
- Process: Boosts amplitude
- Output: Strong signal

👉 Important:

It does NOT understand data, it just boosts signal power.

🔴 Key Point (Very Important for Interview)

👉 Amplifier also amplifies noise

This means:

- Signal ↑
- Noise ↑ ❌

So quality may not improve, only strength increases.

What is a Repeater in Networking?

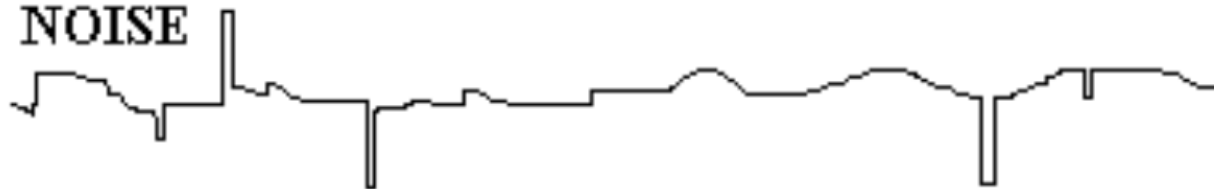
A repeater is a network device that:

👉 Receives a weak signal → regenerates it → sends it forward as a fresh signal

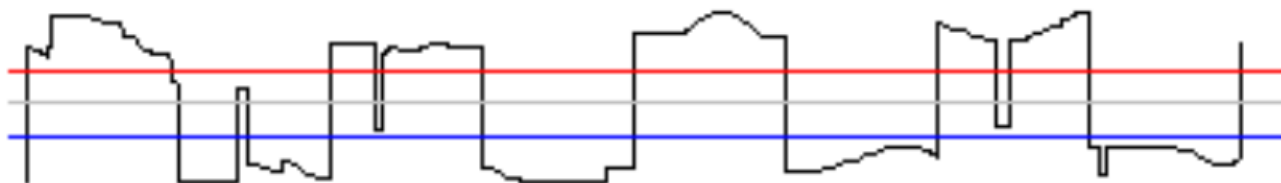
SIGNAL



NOISE



SIGNAL + NOISE





How Repeater Works

Step-by-step:

1. Signal travels → becomes weak (attenuation)
2. Repeater receives it
3. Cleans & regenerates the signal
4. Sends a new strong signal



Why Repeater is Used

- To extend network distance
- To overcome signal loss
- To maintain data integrity

🔴 Key Feature (Important)

👉 Repeater works at:

OSI Layer 1 → Physical Layer

👉 It does NOT:

- Understand IP address
- Filter data
- Make decisions

It only works with **bits** (0s and 1s)

Where Repeater is Used

- Ethernet networks (older LANs)
- Long cable connections
- Fiber optic communication
- Wi-Fi range extenders (modern example)

 Example:

If LAN cable limit = **100 meters**, repeater helps extend beyond that.

Repeater vs Amplifier (Very Important)

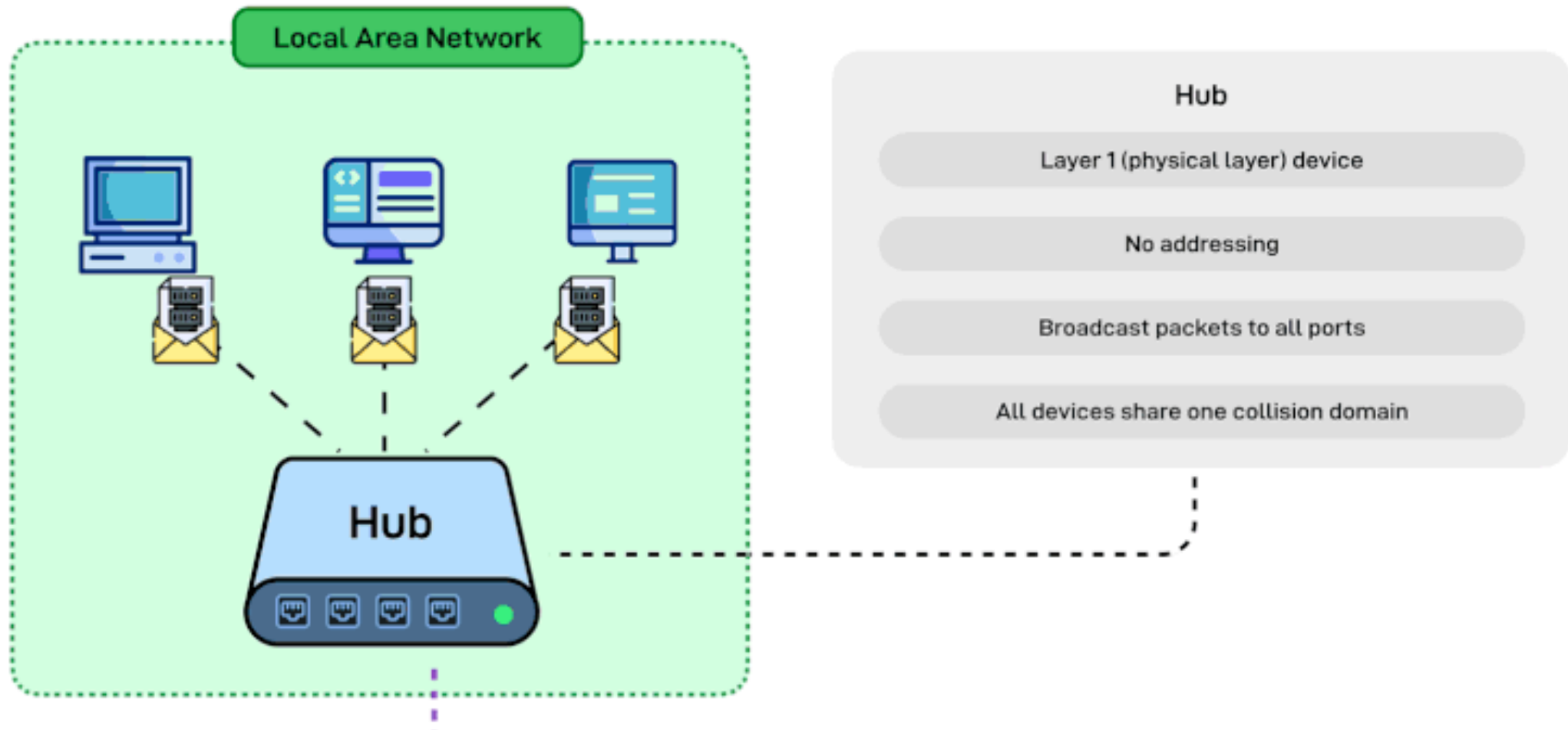
Feature	Repeater	Amplifier
Signal type	Digital	Analog
Function	Regenerates signal ✓	Boosts signal ✓
Noise	Removes noise ✓	Amplifies noise ✗
Intelligence	Basic (clean signal)	No intelligence



What is a Hub in Networking?

A hub is a basic network device that:

👉 connects multiple computers in a LAN and broadcasts data to all devices





How Hub Works

Step-by-step:

1. Device A sends data
2. Hub receives it
3. Hub sends **same data to ALL** ports
4. Every device receives it, but only the correct one accepts

● Key Characteristics (Very Important)

- Works on OSI Layer 1 (Physical Layer)
- No intelligence ✖
- No MAC address table ✖
- No filtering ✖
- Just broadcasts everything

Where Hub is Used

- Old Ethernet networks
- Lab/testing environments
- Almost **obsolete now**

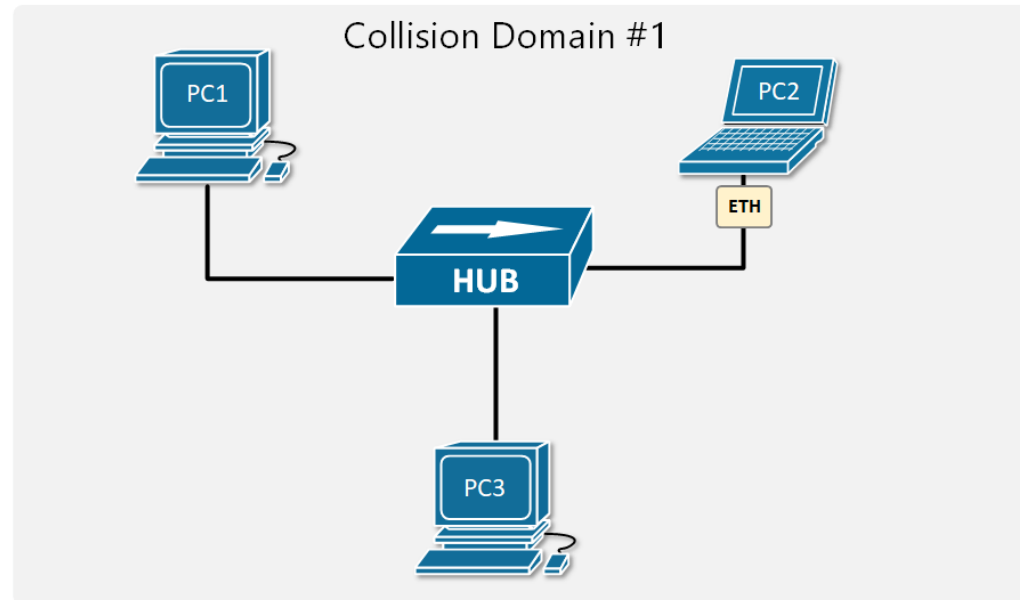
 Modern networks use **switch instead of hub**

💥 Collision Problem

👉 Hub creates **one single collision domain**

Meaning:

- All devices share same bandwidth
- If two devices send data at same time → **collision occurs**





What is a Switch in Networking?

A switch is a network device that:

👉 connects multiple devices in a LAN and sends data only to the intended destination

Step-by-step:

1. Device A sends data to Device B
2. Switch checks **MAC address table**
3. If MAC is known → sends data only to that port 
4. If unknown → temporarily broadcasts (called **flooding**)
5. Learns MAC addresses and stores them

● Key Features (Very Important)

- Works on OSI Layer 2 (Data Link Layer)
 - Uses MAC addresses
 - Reduces collisions ✓
 - Creates **separate collision domains**
 - Faster than hub 🚀
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✨ Collision & Broadcast Domain

- Each port = **separate collision domain** ✓
- Entire switch = **one broadcast domain** (by default)



Where Switch is Used

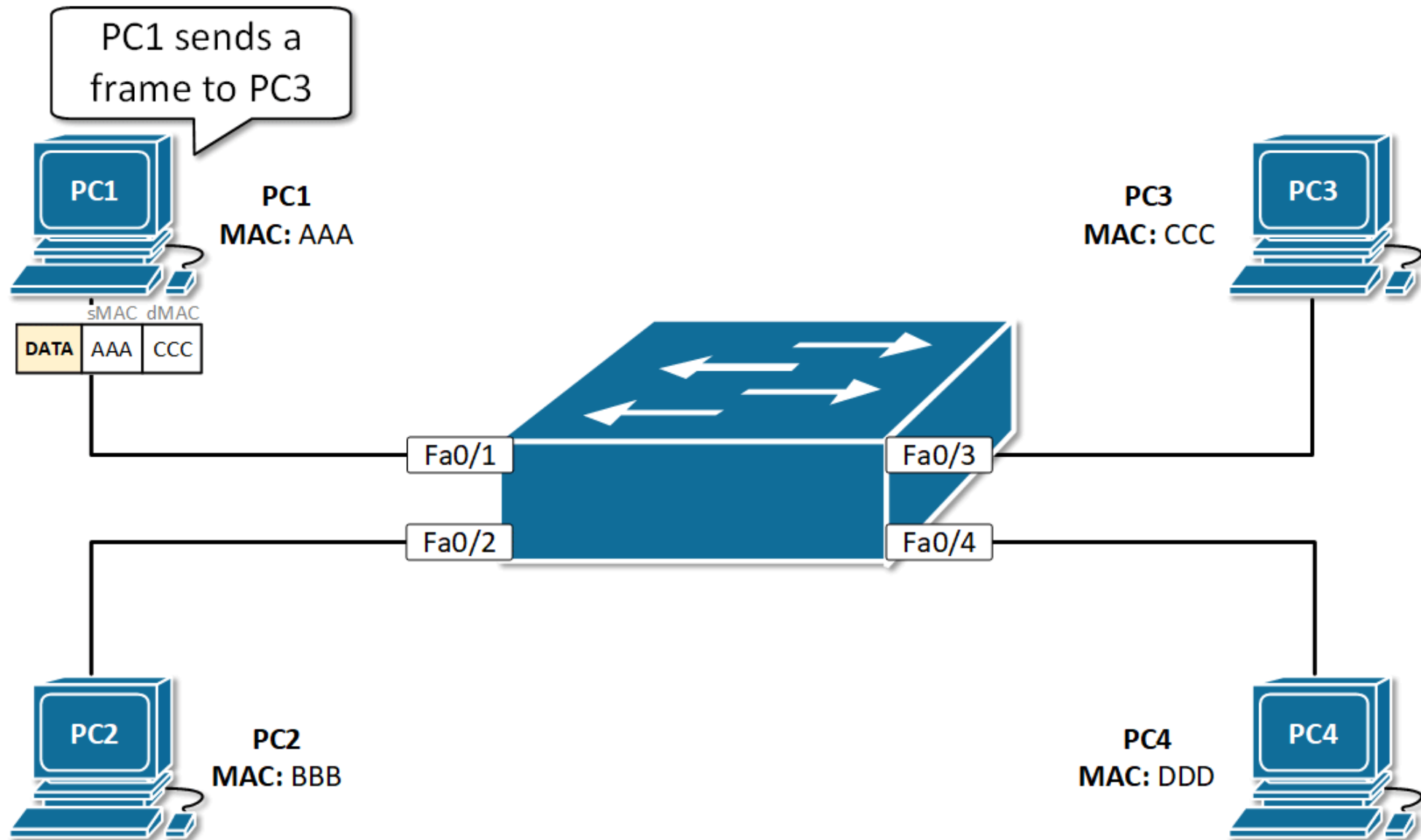
- Office networks
- Data centers
- Home routers (built-in switch)
- Enterprise LANs

MAC Address Table (Core Concept)

- Switch stores:
 - MAC Address
 - Port number

 Example:

MAC Address	Port
A1:B2:C3	Port 1
D4:E5:F6	Port 2



Types of Switch

1. Unmanaged Switch

- Plug and play
- No configuration

2. Managed Switch

- VLAN, security, monitoring
- Used in enterprises

3. Layer 3 Switch

- Can perform routing (IP-based)

Switch vs Hub (Most Asked)

Feature	Switch	Hub
Layer	L2	L1
Data transfer	Unicast (specific)	Broadcast (all)
MAC table	Yes	No
Collision	Low ✔	High ✖
Speed	Fast	Slow

What is a Router?

A router is a network device that:

👉 connects different networks and forwards data using IP addresses





How Router Works

Step-by-step:

1. Device sends data (with **destination IP**)
2. Router receives packet
3. Checks **Routing Table**
4. Decides **best path** (next hop)
5. Forwards packet to destination network

Key Features (Very Important)

- Works on **OSI Layer 3 (Network Layer)**
- Connects **LAN to WAN (Internet)**
- Separates broadcast domains ✓
- Chooses **best path**
- Can apply security (ACL, NAT, etc.)

💣 Broadcast & Collision Logic

- Each router interface =
 - 👉 Separate broadcast domain ✓
 - Collisions:
 - 👉 Not an issue (uses switches internally)
-

⚡ Important Functions

1. Routing

- Static routing
- Dynamic routing (RIP, OSPF, EIGRP)

2. NAT (Network Address Translation)

- Converts **private IP** → **public IP**

👉 Example:

- 192.168.1.10 → Public IP
-

3. Default Gateway

👉 Router acts as:

“Exit point of network”

◆ Routing Table

Router stores:

- Destination network
- Next hop
- Interface

👉 Example:

Network	Next Hop
192.168.1.0	Direct
10.0.0.0	192.168.1.1



Router vs Switch (Important)

Feature	Router	Switch
Layer	L3	L2
Address	IP	MAC
Connects	Network to Network	Devices in same network
Broadcast	Stops broadcast ✓	Forwards broadcast ✗
Decision	Best path routing	MAC-based forwarding

Where Router is Used

- Home Wi-Fi router
 - ISP networks
 - Enterprise networks
 - Internet backbone
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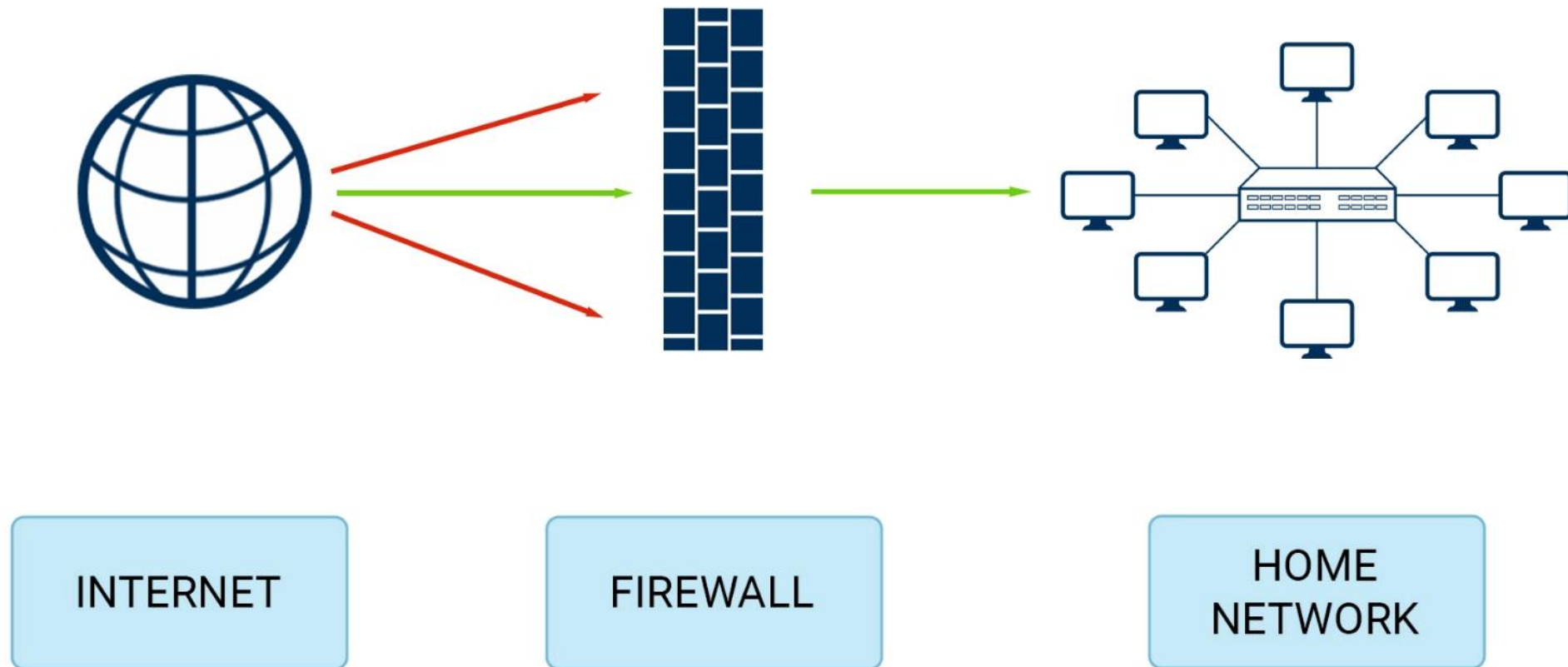
Real-Life Analogy

- **Switch** = Inside same building (rooms connection)
- **Router** = Between cities (road network decision)

🔥 What is a Firewall?

A firewall is a network security device that:

👉 monitors and controls incoming and outgoing traffic based on rules



How Firewall Works

Step-by-step:

1. Data packet comes from internet
2. Firewall checks rules (policies)
3. If allowed → pass 
4. If not allowed → block 



Core Logic of Firewall



Firewall makes decisions based on:

- IP Address
- Port Number
- Protocol (TCP/UDP)
- Application (in advanced firewall)

Key Features (Very Important)

- Protects network from **unauthorized access**
- Filters traffic (Allow / Deny)
- Can log and monitor traffic
- Can prevent attacks (basic level)

Types of Firewall

1. Packet Filtering Firewall

- Checks basic info (IP, port)
 - Fast but less secure
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2. Stateful Firewall

- Tracks connection state
- More secure 

3. Application Layer Firewall

- Works at Layer 7
 - Inspects application data
-

4. Next-Generation Firewall (NGFW)

- Deep inspection
- IDS/IPS features
- Malware protection

Firewall Layers

- Works mainly on:
 - Layer 3 (Network)
 - Layer 4 (Transport)
 - Layer 7 (Application) (advanced)
-

Example (Easy Understanding)

 Rule:

- Allow: Port 80 (HTTP)
- Block: Port 23 (Telnet)

 Result:

- Website opens 
- Telnet blocked 



Firewall vs Router

Feature	Firewall	Router
Purpose	Security	Routing
Decision	Based on rules	Based on path
Focus	Traffic filtering	Packet forwarding
Security	High ✓	Basic

Where Firewall is Used

- Office networks
 - Data centers
 - Cloud environments
 - Home routers (basic firewall built-in)
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Real-Life Analogy

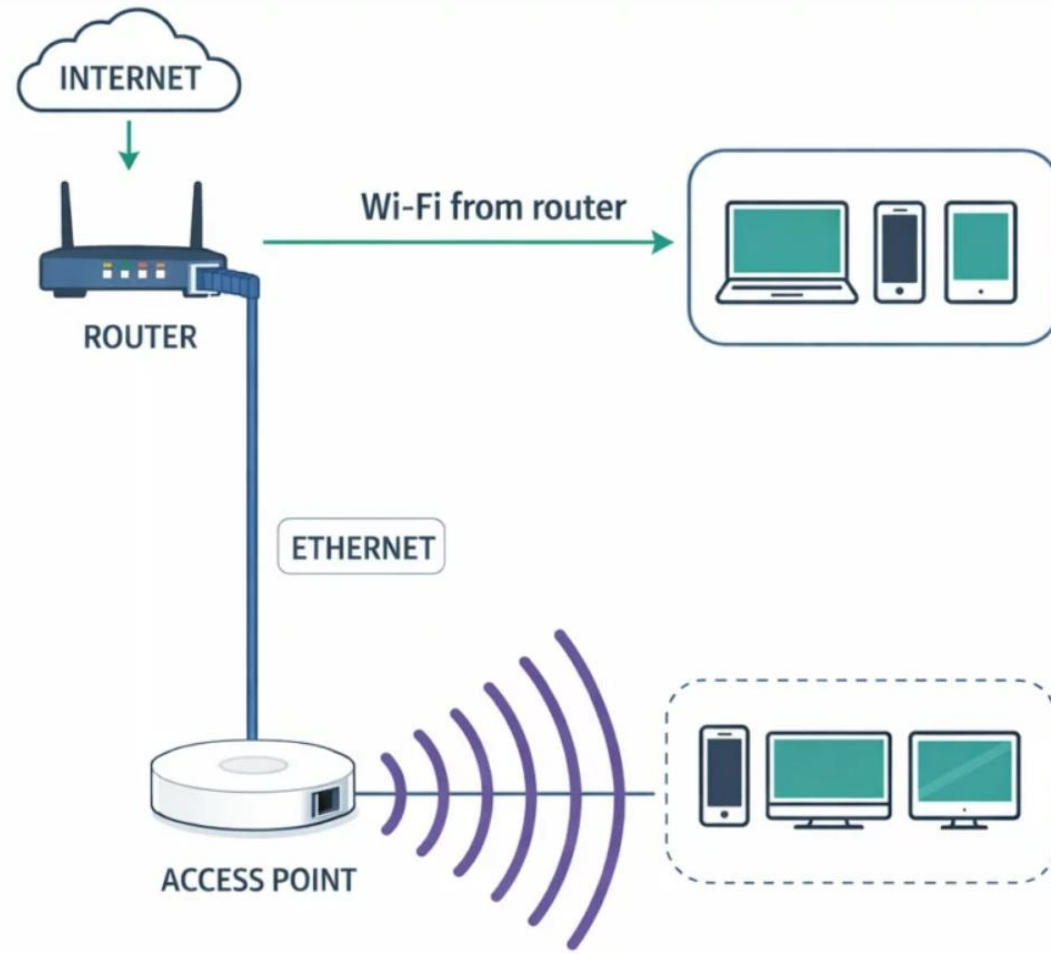
- Firewall = Security guard 
- Router = Traffic manager 



What is an Access Point?

An Access Point (AP) is a network device that:

👉 connects wireless devices (Wi-Fi) to a wired network (LAN)



How Access Point Works

Step-by-step:

1. AP connects to switch/router via cable
2. It creates a **Wi-Fi signal (SSID)**
3. Devices (mobile, laptop) connect wirelessly
4. AP converts:
 - Wireless signal → Wired signal
 - Wired → Wireless

Key Features (Very Important)

- Works on OSI Layer 2 (Data Link Layer)
- Uses MAC addresses
- Provides wireless connectivity
- Supports multiple devices
- Can extend network coverage

Core Logic

 Access Point acts as a **bridge** between:

- Wireless network (Wi-Fi)
- Wired network (Ethernet)

Important Interview Point

 AP does NOT:

- Perform routing
- Assign IP (unless integrated device)

Real-Life Example

 In office:

- Router → connects to internet
 - Switch → connects devices
 - Access Point → provides Wi-Fi
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Where Access Points are Used

- Offices
- Colleges
- Airports
- Homes (Wi-Fi routers have built-in AP)

Types of Access Point

1. Standalone AP

- Used in small networks
 - Works independently
-

2. Wireless Controller-Based AP

- Managed centrally
- Used in enterprises

3. Mesh Access Point

- Multiple APs connected wirelessly
- Large coverage area



SSID (Important Term)



SSID = Network name

Example:

- "Home_WiFi"
- "Office_Network"

Access Point vs Router (Common Confusion)

Feature	Access Point	Router
Function	Provides Wi-Fi	Connects networks
IP Routing	No ❌	Yes ✅
Role	Bridge	Gateway
Internet sharing	Through router	Direct

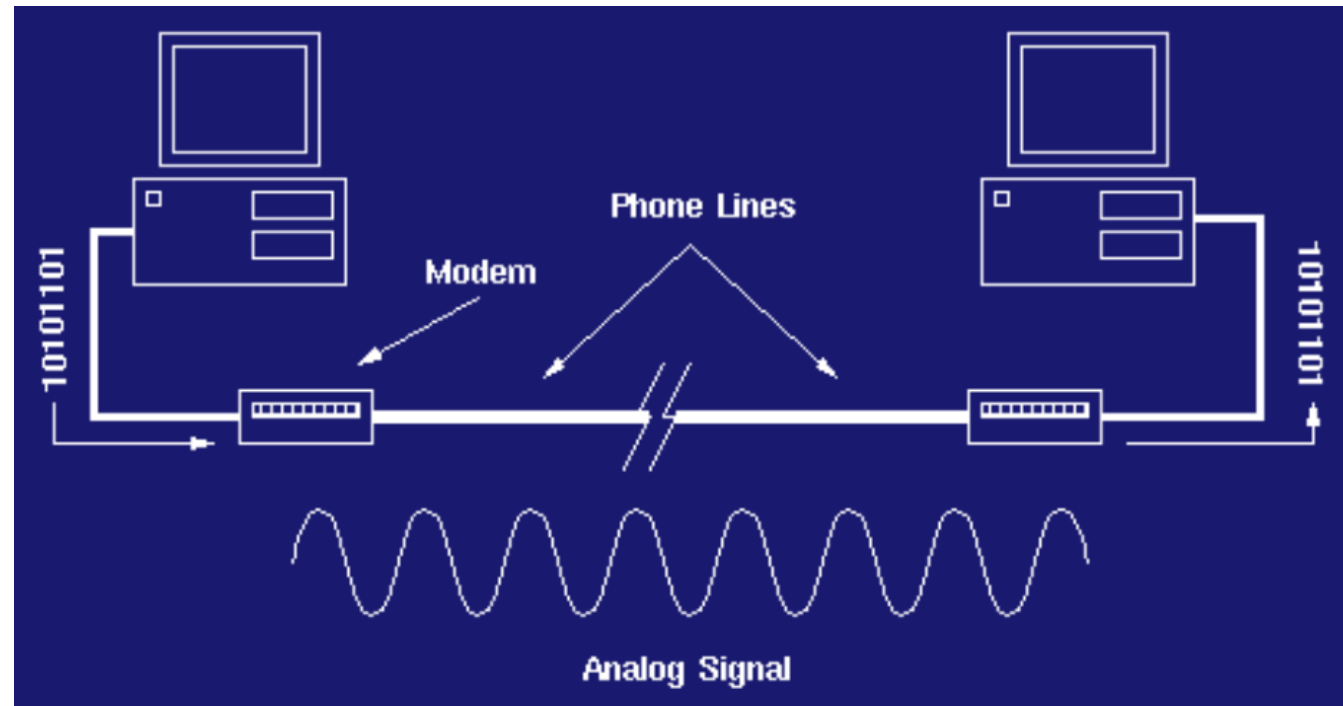
What is a Modem?

A modem is a network device that:

👉 converts digital signals into analog and vice versa

Full form:

MOdulator + DEModulator



How Modem Works

Step-by-step:

1. Computer generates **digital data** (0s and 1s)
2. Modem converts it into **analog signal** (for transmission)
3. Signal travels through ISP line (cable/telephone/fiber)
4. At receiving end → modem converts back to **digital**

● Key Features (Very Important)

- Works on **Physical Layer (Layer 1)**
- Converts:
 - Digital → Analog (Modulation)
 - Analog → Digital (Demodulation)
- Required for **internet connectivity**

⚡ Types of Modem

1. DSL Modem

- Uses telephone line

2. Cable Modem

- Uses coaxial cable (TV cable)

3. Fiber Modem (ONT)

- Uses fiber optic

4. Wireless Modem

- Uses SIM (4G/5G)

